Install Linux On Your Machine

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Topics

- **Choosing your distro**
  Link: `https://youtu.be/8yVlJEzq2eg?si=SWcgh_Bw51Li3lv6`
- **Install your Linux system**
  Link: `https://www.youtube.com/watch?v=C-a5IamFIuM`
- **Switch to Linux while keeping data from Windows / Dual-booting with Windows**
  Link: `https://youtu.be/XevGf0_vQJQ?si=0BqaM8UlUeH1wjdQ`
- **VirtualBox VM setup**
  Link: `https://youtu.be/wX75Z-4MEoM?si=raCzHOzYJy-GZi9S`
- **Post-install (drivers – firewall – backup and restore system)**
  Link: `https://mmbesar.github.io/Tutorials/Ubuntu-24.04-Post/`

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1. Choosing Your Linux Distribution

Before installing Linux, you must pick the right **distro** (distribution). Think of distros like different flavors of the same operating system — they all share the Linux kernel but differ in user interface, package managers, performance, and philosophy.

1.1 What You Should Look For

When choosing a distro, consider:

Distro Selection Guide:

- **Ease of use** — Beginner-friendly UI, simple updates

  - Good Choices: Ubuntu, Linux Mint, Pop!_OS
- **Stability** — Fewer issues & long-term support
  - Good Choices: Ubuntu LTS, Debian
- **Performance** — Works well on older hardware
  - Good Choices: Xubuntu, Linux Lite
- **Security** — Good by default, strong community
  - Good Choices: Fedora, Ubuntu
- **Software availability** — Access to apps & drivers
  - Good Choices: Ubuntu, Fedora

1.2 Recommended Distro for Beginners

Ubuntu 24.04 LTS
Reason: stable, popular, lots of tutorials, works on almost all laptops.

1.3 Quick Tips

- If your PC is older than **10 years**, pick **Xubuntu** or **Linux Mint XFCE** or **MX-Linux** or **Zorin OS**.
- If you're into **gaming**, choose **Pop!_OS** or **Nobara OS**.
- If you want a macOS-like UI, choose **elementaryOS**.
- Avoid Arch-based systems as your first Linux unless you're ready for deeper learning.

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2. Installing Your Linux System

In This section we will use Ubuntu 24.04 for Explanation.

2.1 Before You Install – VERY Important

A. Backup Your Files

The installer will delete the Windows partition if you're switching fully. Backup all files from:

- Desktop
- Documents
- Pictures
- Music
- Videos

Move them to:

- D:\(or another partition)
- External HDD
- USB stick

B. Check Your Boot Mode

Open Start → type **System Information** → look for:

- **UEFI** → modern, secure boot
- **Legacy BIOS** → older systems

This affects partitioning and bootloader behavior.

C. Download the ISO

- Ubuntu official website
- Size [5 - 6]GB

D. Create a Bootable USB

Recommended tool: **Ventoy**

- Supports multiple ISOs on the same flash drive
- Very beginner-friendly

2.2 Try Ubuntu Before Installing

Boot from the USB and choose:

Try Ubuntu (Live Mode)

You can test:

- Wi-Fi
- Bluetooth
- Sound
- Display
- Keyboard
- Touchpad

If everything works → safe to install.

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3. Installing Linux (Manual Partitioning)

This is the *critical* step — mistakes here delete data.

3.1 Choose Manual Installation

DO NOT choose:

- “Erase disk and install Ubuntu”
- “Install alongside Windows” (unpredictable for beginners)

Instead, choose: **Something Else (Manual Partitioning)**

3.2 Identify Windows Partitions to Delete

In the live session, open a terminal:

```
$ lsblk
```

You will see all disks:

- Windows system partitions
- Data partitions
- USB Ventoy

- Empty space

Delete only:

- The Windows partition (C drive)
- Windows system reserved partitions

Never delete your data partitions.

3.3 Creating Linux Partitions

Recommended layout:

Linux Partition Layout:

- **EFI/System** — 512–1024 MB — FAT32 — `/boot/efi` — Needed only for UEFI
- **ROOT** — 40–100 GB+ — EXT4 — / — Main system
- **SWAP** — Optional (2–4 GB) — swap — Only if RAM < 8GB
- **Data (Windows)** — Keep — NTFS — `/mnt/data1` — Do *not* format

Beginners Tip: Don't create a separate `/home` partition. You can add it later when you're more experienced.

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4. Switching to Linux While Keeping Data (Dual Boot)

4.1 How Dual Boot Works

Your disk will contain:

- Windows NTFS partitions
- Linux EXT4 partitions
- ESP (bootloader)
- Shared data partitions

GRUB bootloader will let you choose the OS at startup.

4.2 Before Dual Booting

- Clean Windows (delete temp, uninstall unused programs)
- Run: `chkdsk /f`
- Disable Fast Startup in Windows
- Shrink Windows partition using *Disk Management*

4.3 Partition Strategy

You will create:

- EFI partition (use existing one)
- Root partition
- (Optional) Swap partition

Mount Windows data partitions as:

```
/mnt/data1
/mnt/data2
```

Never check the "format" box.

4.4 Accessing Windows Files After Installation

Open **Files → Other Locations → Ubuntu → mnt**.
Your data partitions appear there.
You can:

- Pin them
- Bookmark them
- Rename them

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5. Installing Linux in VirtualBox (VM Setup)

For students who don't want to touch their main system.

5.1 Why a Virtual Machine?

- Practice Linux commands safely
- Test configurations
- Use Linux tools on a Windows/macOS system
- No risk of data loss

5.2 Basic Requirements

- CPU: Dual-Core
- RAM: 6–8 GB system → allocate 3–4 GB to VM
- Disk: 25–40 GB virtual disk
- GPU: Enable 3D Acceleration

5.3 Recommended VirtualBox Settings

- **Chipset:** ICH9
- **EFI:** Enabled for Ubuntu 24.04
- **CPU:** 2 cores
- **Display:** 128 MB VRAM
- **Network:** Bridged Adapter (for networking labs)

5.4 Installing Guest Additions

Inside VM → Devices → Insert Guest Additions CD
Run:

```
$ sudo apt install build-essential dkms linux-headers-$(uname -r)
$ sudo sh /media/username/VBox*/VBoxLinuxAdditions.run
```

This enables:

- Auto-resize display
- Clipboard sharing
- Drag & drop
- Better performance

6. Post-Installation Setup (Ubuntu 24.04)

6.1 Update System

```
$ sudo apt update && sudo apt upgrade -y
```

6.2 Install Ubuntu Restricted Extras

Adds codecs for:

- MP3
- MP4
- MOV
- Fonts

```
$ sudo apt install ubuntu-restricted-extras
```

6.3 Add Flatpak + Flathub

```
$ sudo apt install flatpak
$ flatpak remote-add --if-not-exists flathub
https://flathub.org/repo/flathub.flatpakrepo
```

Install Flatpak support in GNOME Software:

```
$ sudo apt install gnome-software gnome-software-plugin-flatpak
```

6.4 AppImage Support

```
$ sudo apt install libfuse2
```

6.5 GNOME Extensions & Tweaks

```
$ sudo apt install chrome-gnome-shell gnome-shell-extension-manager
gnome-tweaks
```

6.6 Minimize on Click

Classic:

```
$ gsettings set org.gnome.shell.extensions.dash-to-dock click-action
'minimize'
```

Newer behavior:

```
$ gsettings set org.gnome.shell.extensions.dash-to-dock click-action
'focus-minimize-or-appspread'
```

6.7 Enable New Document Menu

```
$ touch ~/Templates/new
```

6.8 Keyboard Layout Switching

Set:

```
$ gsettings set org.gnome.desktop.wm.keybindings switch-input-source
"['<Shift>Alt_L']"
$ gsettings set org.gnome.desktop.wm.keybindings
switch-input-source-backward "['<Alt>Shift_L']"
```

6.9 System Backups

Timeshift (System Snapshots)

```
$ sudo apt install timeshift
```

Déjà Dup (User files backup)

Built into Ubuntu.

6.10 Final System Update

```
$ sudo apt update && sudo apt upgrade -y && flatpak update -y && sudo
snap refresh
```

Conclusion

By the end of this session, students should be able to:

- ✔ Choose a Linux distro that fits their hardware and needs
- ✔ Install Linux safely and correctly
- ✔ Dual-boot without losing their Windows files
- ✔ Use Linux inside a Virtual Machine
- ✔ Apply essential post-install steps for performance, security, and usability

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